Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS A and B on this page.

PART A

$g(x) = \int_0^x f(t)\,dt$

$g'(x) = f(x)$

$g'(8) = f(8)$

$\boxed{g'(8) = 1}$

$g'(8) = 1$ because $g'(x) = f(x)$. When $x = 8$, $f(8) = 1$ and because the derivative of $g(x)$ equals $f(x)$, $g'(8)$ will also equal 1.

PART B $g''(x) = f'(x)$

$\boxed{\text{POI: } x = 0, 6}$

Because $g''(x) = f'(x)$, when the slope of $f(x)$ changes signs, $g(x)$ will have a point of inflection.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS C and D on this page.

PART C

$g(x) = \int_6^x f(t)\,dt$

$g(12) = \int_6^{12} f(t)\,dt$

$\boxed{g(12) = 18}$

$g(0) = \int_6^0 f(t)\,dt = -\int_0^6 f(t)\,dt$

$g(0) = -\frac{1}{2}\pi(3)^2$

$\boxed{g(0) = -\frac{9\pi}{2}}$

PART D

$g'(x) = f(x)$

$g(6) = 0$

$g(-6) = 0$

$g(12) = 18$

g attains an absolute maximum at $x = 12$ because $g(x) = \int_6^x f(t)\,dt$, and at $x = 6$, the integral equals 12, so $g(6) = 12$.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.